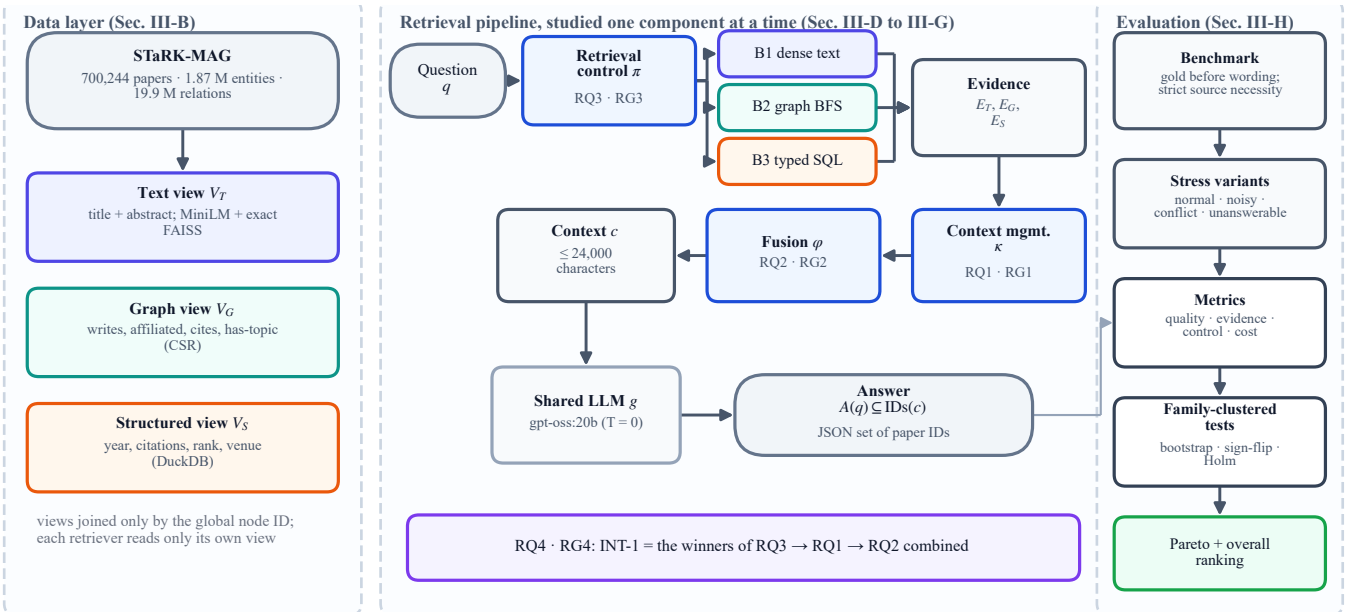

Data layer (Sec. III-B)
STaRK-MAG
700,244 papers · 1.87 M entities · 19.9 M relations
Text view V_T
title + abstract; MiniLM + exact FAISS
Graph view V_G
writes, affiliated, cites, has-topic (CSR)
Structured view V_S
year, citations, rank, venue (DuckDB)
views joined only by the global node ID; each retriever reads only its own view
Retrieval pipeline, studied one component at a time (Sec. III-D to III-G)
Question q
Retrieval control π
RQ3 · RG3
B1 dense text
B2 graph BFS
B3 typed SQL
Evidence
E_T, E_G, E_S
Context mgmt. κ
RQ1 · RG1
Fusion φ
RQ2 · RG2
Context c
$\leq$ 24,000 characters
Shared LLM g
gpt-oss:20b (T = 0)
Answer
$A(q) \subseteq \mathrm{IDs}(c)$
JSON set of paper IDs
RQ4 · RG4: INT-1 = the winners of RQ3 → RQ1 → RQ2 combined
Evaluation (Sec. III-H)
Benchmark
gold before wording; strict source necessity
Stress variants
normal · noisy · conflict · unanswerable
Metrics
quality · evidence · control · cost
Family-clustered tests
bootstrap · sign-flip · Holm
Pareto + overall ranking